Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

Proc IEEE Int Symp Biomed Imaging. Author manuscript; available in PMC 2025 August 13.

Published in final edited form as:

Proc IEEE Int Symp Biomed Imaging. 2025 April ; 2025: . doi:10.1109/isbi60581.2025.10981128.

OVERALL SURVIVAL PREDICTION OF BRAIN TUMOR PATIENTS WITH MULTIMODAL MRI USING SWIN UNETR

Gihyeon Kim[1,2], **Fangxu Xing**[1], **Hyoun-Joong Kong**[1,3], **Emiliano Santarnecchi**[1], **Helen A. Shih**[1], **Thomas Bortfeld**[1], **Georges El Fakhri**[4], **Xiaofeng Liu**[4], **Jang-Hwan Choi**[2], **Jonghye Woo**[1]

[1]Massachusetts General Hospital and Harvard Medical School, Boston, MA, USA

[2]Ewha Womans University, Seoul, Korea

[3]Seoul National University Hospital and Seoul National University College of Medicine, Seoul, Korea

[4]Yale University, New Haven, CT, USA

Abstract

Accurate prediction of glioblastoma patient survival can significantly aid in personalized treatment planning. While pre-operative multimodal magnetic resonance imaging (MRI) offers complementary information, current methods are constrained by relatively limited data and largely rely on hand-crafted features extracted from segmentation results. To address these issues, in this work, we propose a data-efficient multi-task framework to take advantage of hierarchical segmentation features within advanced Swin UNETR for survival prediction. By integrating multi-scale features, we are able to capture detailed spatial information and global context, while employing the shifted window mechanism to maintain computational efficiency and scalability for 3D volumes. We further alleviate survival data scarcity through segmentation pre-training, while the features are fine-tuned to align with the survival prediction task and refined by statistical F-values. In addition, age information is incorporated alongside the extracted features to enhance survival prediction performance. Through comprehensive evaluations on the BraTS dataset, we demonstrate that our model achieves superior segmentation accuracy and state-of-the-art survival prediction performance, offering a robust solution for clinical prognosis in glioblastoma patients.

Keywords

Survival Prediction; Multi-task Learning; Hierarchical Feature; Multimodal MRI; Deep Learning

1. INTRODUCTION

Glioblastoma (GBM) carries one of the worst prognosis among human cancers with a median survival time of 15 months after initial treatment [1]. Accurate survival prediction

6.COMPLIANCE WITH ETHICAL STANDARDS

6.This research study was conducted retrospectively using human subject data made available in open access. Ethical approval was not required as confirmed by the license attached with the open-access data.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

plays a crucial role in the diagnosis, treatment planning, and prognosis of brain tumors. Multimodal magnetic resonance imaging (MRI), including T1-weighted, T1ce (contrast enhanced), T2-weighted, and FLAIR (FLuid-Attenuated Inversion Recovery) MRI, provides complementary information about tumors that is essential for both clinical practice and scientific research, aiding in this prediction task.

In recent years, a variety of deep learning (DL) models have been proposed to identify suitable representations of high-dimensional and complex data for various tasks, including segmentation and prediction. For accurate survival prediction, segmentation plays a critical role, by providing essential tumor-specific features, such as size, shape, and location, which directly impact survival outcomes. By jointly training models for both segmentation and prediction, these approaches can significantly improve survival prediction accuracy by leveraging the detailed spatial information obtained from segmented tumor regions.

Many previous survival prediction methods, however, relied on hand-crafted features, such as radiomics or volumetric features, derived from segmentation results [2, 3, 4]. Recently, DL-based approaches have demonstrated superior performance by jointly automating brain tumor segmentation and using the extracted tumor-specific features to improve the accuracy of survival prediction. For example, [5] extracted deep features from the CNN-based model with a context encoding module and applied linear regression. Multi-task learning methods exploring the correlation between brain tumor segmentation and overall survival have also been proposed. [6] ensembled the prediction results from the global and local branch which derives from the survival prediction and segmentation modules, respectively. [7] proposed a method that shares the same encoder for segmentation and survival prediction.

In addition, transformer-based methods have seen significant development in medical image analysis and have been applied to survival prediction in brain tumors. For instance, several works [7, 8, 9] adapted a Vision Transformer (ViT) for feature extraction from 3D MRI scans, where the extracted features are aggregated and passed through fully connected layers to predict survival. The self-attention mechanism in the ViT allows the model to capture both local and global context by identifying relationships between different image patches. Similarly, [10] utilized features from the Swin Transformer UNet encoder to capture both local and global features for survival prediction. Moreover, [11] proposed a survival regression network that is jointly trained with a pretrained segmentation backbone based on UNETR++. The hierarchical structure of UNETR++ enables effective aggregation of low-level and high-level features, enhancing the accuracy of survival prediction.

Despite these advances, previous DL methods have not been efficiently applied to survival prediction, largely due to the limited availability of training data. For instance, models that participated in the Multimodal Brain Tumor Segmentation (BraTS) challenge trained survival prediction networks with fewer than 250 patients. In addition, many recently developed DL models focus solely on segmentation tasks, without accounting for the correlation between tumor characteristics and survival days. Furthermore, previous approaches typically relied on features extracted from the encoder, bottleneck, or predicted segmentation mask of the segmentation network for overall survival prediction, potentially

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

limiting their ability to make use of the complex relationship between tumor features and patient survival.

To address the aforementioned limitations, we aim to develop a DL model to predict patient survival days from preoperative, multimodal MR scans, utilizing automated brain tumor segmentation. We leverage a segmentation backbone pre-trained on a large-scale 3D MRI dataset to obtain rich feature representations, effectively mitigating the challenges posed by a small dataset. We integrate multi-scale imaging features with clinical data to enhance overall survival prediction. By incorporating features from both the encoder and decoder, our model captures both global context and fine details, providing a more comprehensive representation of the images. Furthermore, we utilize shifted windows in the swin transformer to ensure computational efficiency when processing high-resolution medical images. We detail our methods in the following section.

2. METHODS

In GBM survival prediction using multiparametric MRI, we are provided with paired data $\{x, y\}$, where $x \in \mathbb{R}^{H \times W \times D \times 4}$ represents registered multimodal MRI data and y denotes the corresponding survival days, where H, W, and D are height, width, and slice of 3D MRI volume, respectively. The objective is to train a model $\mathcal{F}: x \rightarrow y$ that accurately maps MRI features to patient survival duration. Extracting meaningful feature representations from complex multimodal 3D MRI data remains challenging, especially with limited datasets. In this work, we introduce a data-efficient, multi-task framework that leverages the hierarchical segmentation features within the advanced Swin UNETR architecture to enhance survival prediction accuracy.

2.1. Unified Multi-task Framework

The architecture of our proposed framework is illustrated in Fig. 1. The network is trained on the four concatenated MRI modalities. Specifically, we used Swin UNETR as the segmentation backbone to label tumor sub-regions. Although our multi-task segmentation and survival prediction framework can accommodate any U-shaped 3D segmentation backbone, we selected the pre-trained weights of Swin UNETR. This model was trained on 1,251 multi-modal 3D MRI volumes from the BraTS 2021 dataset [12] and demonstrated good performance on the validation dataset. Among the five models from the five-fold data splits, we selected the model with a mean Dice score of 89.24, as it showed the best survival prediction results. We then fine-tuned this Swin UNETR model using the pre-trained weights to fit our relatively small dataset for joint segmentation and survival prediction.

2.2. Hierarchical Feature Extraction

The features related to tumor sub-regions were extracted from the intermediate layers of the fine-tuned Swin UNETR as in Fig. 2. We utilized the output of residual blocks in the decoder layers. These features combine both skip connections from the encoder and upsampled decoder features. They are further refined by residual connections consisting of convolutional layers with a kernel size of 1×1×1 and instance normalization. Deep features

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

derived from multi-scale residual blocks are fed into a global average pooling layer. A total of 768 features are derived from five residual blocks. The features related to overall survival were selected using SelectKBest, which applies a statistical F-value between labels and features [13]. The optimal number of features was selected based on results across all folds.

2.3. Survival Prediction

A total of 45 features related to tumor sub-regions were selected from the segmentation network. According to previous studies [3, 4], patient age shows a strong correlation with overall survival in brain tumor cases. Therefore, we combined the clinical feature 'age' with the features extracted from Swin UNETR. We tested three linear regression models, including ridge, lasso, and multivariate linear regression, to capture linear patterns from complex deep features. Lastly, the model's predictions are clipped at a minimum value of zero to ensure non-negative survival day estimates.

3. EXPERIMENTS

We utilized the open dataset from the BraTS challenge for brain tumor segmentation and overall survival prediction. Since ground truth segmentation masks were only available in the training data, we expanded our dataset by including patient cases added to the BraTS 2020 dataset. We used all patients with available overall survival data, resulting in 210 cases from BraTS 2019 and 25 cases from BraTS 2020.

Overall survival days were classified into three categories: short-term (below 10 months), mid-term (between 10-15 months), and long-term (more than 15 months). Following the challenge guidelines, only patients with gross total resection (GTR), referring to complete removal of all visible tumor tissue, were evaluated for survival prediction. The patient cohort had a mean age of 60.75 years with a standard deviation of 11.89 years. Additional clinical information is summarized in Table 1.

3.1. Preprocessing

All images were co-registered to the same anatomical template, resampled to 1 mm^3 resolution, and skull-stripped following the protocol in [14]. The images were then standardized into a Z-distribution to have zero mean and unit standard deviation. Various data augmentation techniques were applied to improve the robustness and generalization ability of our model. Specifically, the original volume of size of 240×240×155 was randomly cropped to 128×128×128, followed by random flipping. We also simulated image variations through random intensity normalization, scaling (range: [0.9, 1]), and shifting (range: [−0.1, 0.1]).

Patient data were provided in four different MR image modalities: T1, T1ce, T2, and T2-FLAIR. The tumor ground truth was determined and verified by an expert neuroradiologist. These annotations included three tumor subtypes—necrotic/non-enhancing tumor core, enhancing tumor, and peritumoral edema. The annotations were combined into three sub-regions to evaluate segmentation performance: Tumor Core (TC), Enhancing Tumor (ET), and Whole Tumor (WT).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

3.2. Experimental Settings

The dataset was divided using stratified five-fold cross-validation based on the three overall survival categories. The proposed framework was iterated five times, with a 60:20:20 ratio for training, validation, and test sets, respectively. For the segmentation network, evaluation was based on the best validation average Dice score across three tumor sub-regions. The segmentation backbone used a learning rate of 1e-6 and a weight decay of 1e-5 with an Adam optimizer. The batch size was set to 1, and the model was trained for 100 epochs using a cosine annealing learning rate scheduler. We used a sliding window approach with 0.6 overlap to ensure inference was available for the whole image regardless of size. To prevent overfitting, a dropout rate of 0.2 was applied within the multilayer perceptron (MLP) layers of the swin transformer blocks in the encoder. For the survival prediction, the alpha value that controls the strength of regularization were experimented with [0.01, 0.1, 1, 10, 100].

3.3. Evaluation

All predictions were concatenated and evaluated based on the patients with gross total resection for survival prediction. The evaluation metrics include accuracy for classification into three survival days categories, medianSE, stdSE, MSE, MAE, and spearman rank correlation for the regression task.

4. RESULTS AND DISCUSSION

The performance analysis was carried out on the segmentation backbone, extracted features, and survival prediction model. A 3D U-Net-based model with similar dimensions to Swin UNETR was employed as the baseline for the segmentation network. We used a modified 3D U-Net because generic U-Net architectures with minor modifications have shown good performance for segmentation tasks [15].

4.1. Comparison of Segmentation Network

The segmentation performance averaged across five folds is presented in Table 2. To align with the Swin UNETR architecture, we applied residual connections to every convolutional block of the 3D U-Net model. A dropout rate of 0.2 was used between the encoder layers to ensure stable learning. We trained the residual U-Net with the same settings but with a slightly higher learning rate of 5e-5. The results show that Swin UNETR outperformed the baseline. Specifically, it achieved higher performance in segmenting internal structures, tumor core, and enhancing tumor. We attribute this to the strong feature extraction capability of the transformer-based architecture in the encoder blocks.

4.2. Overall Survival Prediction

We compared three regression models and achieved the highest performance using ridge regression with an alpha value of 100 as shown in Table 3. The L2 penalty effectively managed feature correlations from deep features and reduced overfitting. The low stdSE of our method indicates stable model performance, while the high SpearmanR correlation suggests effective capture of non-linear relationships. The proposed method achieved high accuracy for survival day classification, comparable to previous methods [3, 16] from

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

the BraTS challenge, where the highest reported accuracy for survival prediction did not exceed 63%. In addition, the selected 45 features were distributed across all stages of the segmentation network, with more features extracted from deeper layers. Table 4 shows that using features from individual layers without concatenation resulted in lower performance. The results indicate that our proposed framework effectively leverages multi-scale contextual information from the hierarchical encoder of the segmentation backbone.

5. CONCLUSION AND FUTURE WORK

In this work, we presented a novel multi-task framework for GBM patient survival prediction using pre-operative multimodal MRI data. Our framework integrates multi-scale segmentation features with clinical information, patient age, while maintaining computational efficiency through shifted window attention mechanisms. Experimental results on the BraTS dataset demonstrate that our method achieves state-of-the-art performance in both tumor segmentation and survival prediction tasks. The strong performance can be attributed to three key aspects: (1) effective use of pre-trained models to address data scarcity, (2) integration of multi-scale features to capture both local and global context, and (3) feature selection guided by statistical measures. Future work includes the incorporation of additional clinical features and the application of this framework to other types of tumors. Our framework provides a promising step toward accurate prognostic tools for clinical decision-making in GBM treatment.

ACKNOWLEDGMENTS

This work is partly supported by NIH R01CA290745 and NIH R21EB034911.

8. REFERENCES

[1]. Chen Jian, McKay Renée M, and Parada Luis F, “Malignant glioma: lessons from genomics, mouse models, and stem cells,” Cell, vol. 149, no. 1, pp. 36–47, 2012. [PubMed: 22464322]

[2]. Wang Shuo, Dai Chengliang, Mo Yuanhan, Angelini Elsa, Guo Yike, and Bai Wenjia, “Automatic brain tumour segmentation and biophysics-guided survival prediction,” in Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 5th International Workshop, BrainLes 2019. Springer, 2020, pp. 61–72.

[3]. Feng Xue, Tustison Nicholas J, Patel Sohil H, and Meyer Craig H, “Brain tumor segmentation using an ensemble of 3d u-nets and overall survival prediction using radiomic features,” Frontiers in computational neuroscience, vol. 14, pp. 25, 2020. [PubMed: 32322196]

[4]. Sun Li, Zhang Songtao, Chen Hang, and Luo Lin, “Brain tumor segmentation and survival prediction using multimodal mri scans with deep learning,” Frontiers in neuroscience, vol. 13, pp. 810, 2019. [PubMed: 31474816]

[5]. Pei Linmin, Vidyaratne Lasitha, Rahman Md Monibor, and Iftekharuddin Khan M, “Context aware deep learning for brain tumor segmentation, subtype classification, and survival prediction using radiology images,” Scientific Reports, vol. 10, no. 1, pp. 19726, 2020. [PubMed: 33184301]

[6]. Zhao Guojing, Jiang Bowen, Zhang Jianpeng, and Xia Yong, “Segmentation then prediction: a multi-task solution to brain tumor segmentation and survival prediction,” in Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 6th International Workshop, BrainLes 2020, pages=492–502, year=2021, organization=Springer.

[7]. Wu Wenxia, Yan Jing, Zhao Yuanshen, Sun Qiuchang, Zhang Huailing, Cheng Jingliang, Liang Dong, Chen Yinsheng, Zhang Zhenyu, and Li Zhi-Cheng, “Multi-task learning for concurrent

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

survival prediction and semi-supervised segmentation of gliomas in brain mri," Displays, vol. 78, pp. 102402, 2023.

[8]. You Huixia, Zhang Jiankai, Zhao Yuanshen, Mo Tong, Fang Diangang, Lv Xiaofei, Li Zhi-Cheng, Wang Haifeng, Liang Dong, Zeng Hongwu, et al. , "Riskformer: Survival prediction from mr imaging in patients with idh-wildtype glioblastoma," Displays, vol. 79, pp. 102493, 2023.

[9]. Xu Xuan and Prasanna Prateek, "Brain cancer survival prediction on treatment-naïve mri using deep anchor attention learning with vision transformer," in 2022 IEEE 19th International Symposium on Biomedical Imaging (ISBI). IEEE, 2022, pp. 1–5.

[10]. Tran Minh-Trieu, Yang Hyung-Jeong, Kim Soo-Hyung, and Lee Guee-Sang, "Prediction of survival of glioblastoma patients using local spatial relationships and global structure awareness in flair mri brain images," IEEE Access, vol. 11, pp. 37437–37449, 2023.

[11]. Kwon Junmo, Kim Jonghun, and Park Hyunjin, "Leveraging segmentation-guided spatial feature embedding for overall survival prediction in glioblastoma with multimodal magnetic resonance imaging," Computer Methods and Programs in Biomedicine, vol. 255, pp. 108338, 2024. [PubMed: 39042996]

[12]. Hatamizadeh Ali, Nath Vishwesh, Tang Yucheng, Yang Dong, Roth Holger R, and Xu Daguang, "Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images," in International MICCAI brainlesion workshop. Springer, 2021, pp. 272–284.

[13]. Pedregosa Fabian, Varoquaux Gaël, Gramfort Alexandre, Michel Vincent, Thirion Bertrand, Grisel Olivier, Blondel Mathieu, Prettenhofer Peter, Weiss Ron, Dubourg Vincent, et al. , "Scikit-learn: Machine learning in python," the Journal of machine Learning research, vol. 12, pp. 2825–2830, 2011.

[14]. Bakas Spyridon, Akbari Hamed, Sotiras Aristeidis, Bilello Michel, Rozycki Martin, Kirby Justin S, Freymann John B, Farahani Keyvan, and Davatzikos Christos, "Advancing the cancer genome atlas glioma mri collections with expert segmentation labels and radiomic features," Scientific data, vol. 4, no. 1, pp. 1–13, 2017.

[15]. Isensee Fabian, Petersen Jens, Klein Andre, Zimmerer David, Jaeger Paul F, Kohl Simon, Wasserthal Jakob, Koehler Gregor, Norajitra Tobias, Wirkert Sebastian, et al. , "nnu-net: Self-adapting framework for u-net-based medical image segmentation," arXiv preprint arXiv:1809.10486, 2018.

[16]. Agravat Rupal R and Raval Mehul S, "A survey and analysis on automated glioma brain tumor segmentation and overall patient survival prediction," Archives of Computational Methods in Engineering, vol. 28, no. 5, pp. 4117–4152, 2021.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

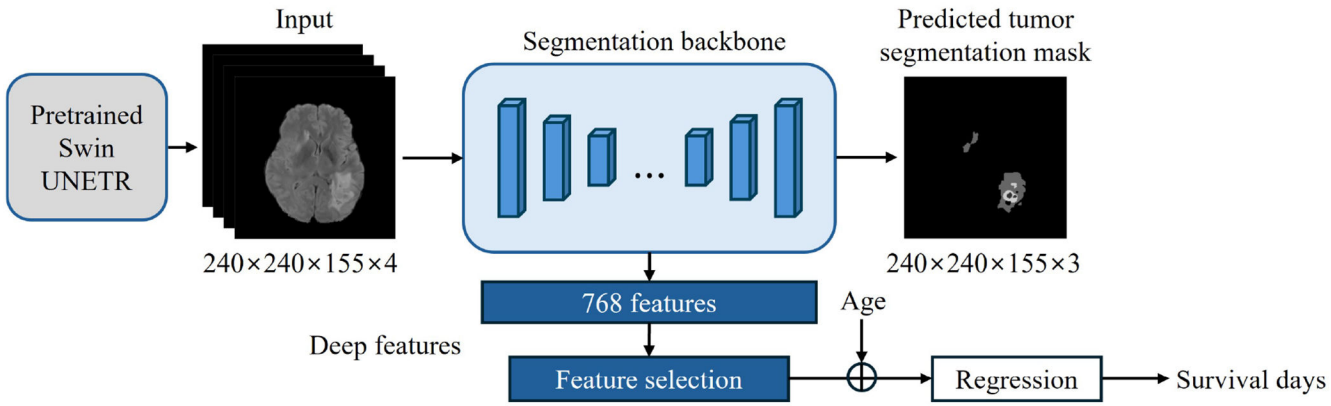


Fig. 1.

Overview of our unified multi-task framework for GBM survival prediction, which exploits hierarchical features from a pre-trained Transformer backbone and fine-tunes the pre-trained Swin UNETR segmentation model through multi-task learning.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

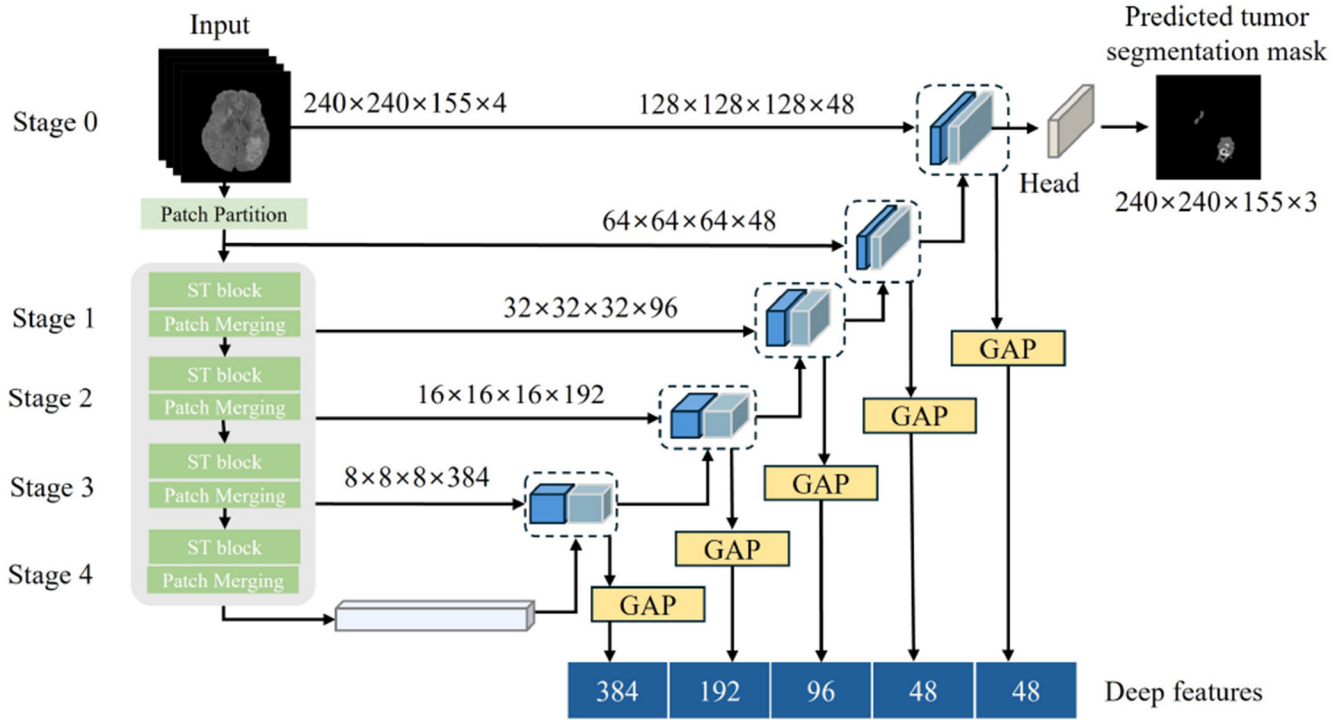


Fig. 2.

Workflow of feature extraction from the segmentation backbone. Features from skip connections are combined with upsampled decoder features and fed into a global average pooling layer. The resulting multi-scale features from all model stages are then concatenated.

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 1.

Clinical information from BraTS (2019, 2020) datasets. OS, GTR, and STR denote overall survival, gross total resection, and subtotal resection, respectively.

Resection status	GTR	STR	Unknown	Total
	118	10	107	235
OS category	short-term	mid-term	long-term	
	91	59	85	

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 2.

Five-fold cross-validation benchmarks in terms of mean Dice score values of three tumor sub-regions. TC, ET, and WT denote Tumor Core, Enhancing Tumor, and Whole Tumor, respectively.

Segmentation backbone	Average Dice score		
	TC	ET	WT
Swin UNETR (Ours)	**0.8397**	**0.7931**	0.8221
Residual U-Net (Baseline)	0.7894	0.7721	**0.8484**

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 3.

Comparison of the survival prediction results based on different regression methods and segmentation network. The residual U-Net with ridge regression is used as a baseline.

	Accuracy	medianSE	stdSE	SpearmanR	MSE	MAE
Ridge R (Ours)	**62.71 %**	28,566	293	**0.6149**	85,245	218
LASSO R	61.86 %	29,334	294	0.5924	86,436	214
Linear R	61.86 %	32,554	301	0.5924	90,150	226
Baseline	56.78 %	**22,943**	**270**	0.5382	**72,889**	**207**

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 4.

Comparison of the survival prediction results based on the stage of the extracted features.

	Accuracy	medianSE	stdSE	SpearmanR	MSE	MAE
Total (Stage 0-4)	**62.71 %**	28,566	293	**0.6149**	85,245	218
Stage 4	56.78 %	22,532	**288**	0.5665	**83,214**	**202**
Stage 3	52.54 %	20,210	301	0.4710	90,558	212
Stage 2	53.39 %	**19,149**	299	0.5139	89,035	213
Stage 1	52.54 %	33,023	311	0.4269	96,425	229
Stage 0	50 %	28,788	318	0.3703	100,938	230